

AI Data Center Cooling System Overview

System Boundary and Definitions

Definition: The primary cooling loop is the closed fluid path that carries heat from compute cold plates to a facility heat exchanger.

Definition: Heat rejection is the transfer of collected thermal energy from the facility loop to an external sink without mixing the two fluids.

Process: Stage 1 moves coolant through cold plates where server heat enters the primary loop.

Process: Stage 2 transfers heat across the facility heat exchanger before the cooled fluid returns to the compute row.

Constraint: The synthetic reference design assumes redundant pumps and isolation valves at each service boundary.

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Sensors, Controls, and Operating Band

Process: Temperature, flow, and differential-pressure sensors feed a local controller that adjusts pump speed and valve position every control interval.

Statistic: The synthetic normal supply-temperature target is 27 C with an allowed steady-state band of plus or minus 1 C.

Statistic: The synthetic design target for supply-to-return temperature rise is 8 C at the nominal compute load.

Claim: A loss of reliable flow telemetry can hide a developing restriction until component temperatures begin to rise.

Claim: Redundant sensing reduces single-point blind spots but does not remove calibration drift risk.

Table: The operating-band text region lists supply temperature 26-28 C, temperature rise 7-9 C, and flow availability above 98 percent.

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Operating Sequence and Boundaries

Process: On a high-temperature alarm, the controller verifies sensor agreement, raises pump output, opens the standby path, and requests compute-load reduction if recovery fails.

Claim: The control sequence depends on maintained valve authority and a verified standby path.

Claim: Facility heat-rejection capacity remains a shared upstream constraint even when the primary loop is healthy.

Limitation: This synthetic overview describes normal and first-response operation only; it does not model site-specific pipe geometry or vendor equipment curves.